

Algorithms HW

1. A *circular polygon* $\mathcal{C}$ is a simply connected open set of $\mathbb{C}$ whose boundary is a union of circular arcs and is homeomorphic to a circle. Show that any Riemann map $\Delta \rightarrow \mathcal{C}$ extends to a homeomorphism $\overline{\Delta} \rightarrow \overline{\mathcal{C}}$.

Suppose for contradiction that a Riemann map $\Delta \rightarrow \mathcal{C}$ could not be extended continuously to the boundary. That is, we would have two sequences in Δ converging to some point on the boundary, $z_n, z'_n \rightarrow z_0 \in S^1$, such that their images converged to distinct points on $\partial\mathcal{C}$. In other words, there is some positive δ such that $|f(z_n) - f(z'_m)| > \delta$ for all n, m .

Consider the following series of calculation. Consider circles α_r of radius r about z_0 . For sufficiently large n, m and sufficiently small r there are points in both sequences which lie on α_r . Now, by the fundamental theorem of calculus we have

$$\begin{aligned} \delta < |f(z_n) - f(z'_m)| &= \left| \int_{\alpha_r} f'(\xi) d\xi \right| \\ &= \left| \int_{\theta_1(r)}^{\theta_2(r)} f'(z_0 + re^{i\theta}) r d\theta \right| \\ &\leq \left| \int_{\theta_1(r)}^{\theta_2(r)} \right| |f'(z_0 + re^{i\theta})| r d\theta \end{aligned}$$

where the second equality is given by parameterizing α_r and the final inequality given by the triangle inequality; the angles $\theta_1(r), \theta_2(r)$ are the angles on which z_n, z'_m respectively lie on α_r . Now, squaring both sides and then applying Cauchy-Schwarz (taking $h = f^{1/2}$, $g = r^{1/2}$) gives

$$\begin{aligned} \delta^2 &\leq \int |f'(z)|^2 r d\theta \int_{\theta_1(r)}^{\theta_2(r)} r d\theta \\ &\leq \int |f'(z)|^2 r d\theta \int_0^{2\pi} r d\theta \\ &\leq \int |f'(z)|^2 r d\theta (2\pi r). \end{aligned}$$

Rearranging this inequality and then integrating over r gives

$$\int_{r=0}^R \frac{\delta^2}{r} dr \leq \int_r \int_{\theta} |f'(z)|^2 r d\theta dr = \int \int |f'|^2 dA = \text{Area}(f(\Delta)),$$

where the last equality is given by the fact that the Jacobian for the appropriate transformation is $|f'(z)|^2$. Now finally, notice that the left hand side diverges, but the right hand side is finite, since it is the area of a bounded open set. A contradiction, and so it must have been that f extended continuously to the boundary.

understand this jacobian statement, make a comment about the boundary being nice.

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2. Let $f : \Delta \rightarrow \Delta$ be a holomorphic function that extends continuously to $\bar{\Delta} \rightarrow \bar{\Delta}$, with $f(S^1) \subset S^1$. Show that f can be extended to a holomorphic function on $\mathbb{C}$ with finitely many points removed.

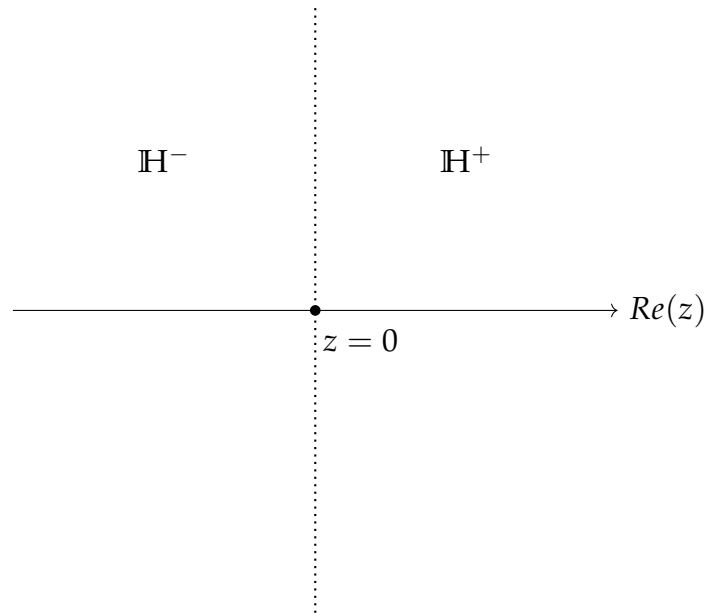
Recall that we have a biholomorphism from the $\Delta \rightarrow \mathbb{H}$ given by

$$h(z) = i \cdot \frac{1+z}{1-z}.$$

This map extends continuously to the boundaries $\bar{\Delta} \rightarrow \bar{\mathbb{H}} \cup \infty$ with $h(1) = \infty$. And so we can define a holomorphic function $g : \mathbb{H} \rightarrow \mathbb{H}$ by $g = h \circ f \circ h^{-1}$ which extends continuously to the actual boundary $\bar{\mathbb{H}} \cup \infty$ and $g(\bar{\mathbb{H}} \cup \infty) \subseteq \bar{\mathbb{H}} \cup \infty$. The preimages of 1 under f are those which get sent to infinity under g .

Suppose for a moment that we have a single point on the real line which gets sent to infinity. Say that we translate our coordinates so that this point is $z = 0$. Let us denote the “right half upper plane” by $\mathbb{H}^+ := \{z \in \mathbb{H} : \operatorname{Re}(z) > 0\}$ and similarly let $\mathbb{H}^-$ denote the “left upper half plane”.

Now, the restriction of g to $\mathbb{H}^+$ defines a holomorphic function which extends continuously to the interval $(0, \infty)$ with the image of $(0, \infty) \subseteq \mathbb{R}$. The Schwarz reflection principle then allows us to extend $g|_{\mathbb{H}^+}$ to a holomorphic function on the entire right half plane $\mathbb{R}^+ := \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$. The same argument means that we can extend $g|_{\mathbb{H}^-}$ to a holomorphic function on the entire left half plane. We now have a holomorphic function $\mathbb{C} \setminus L \rightarrow \mathbb{C}$ where L is the vertical line through $z = 0$. Let us denote this function by $\tilde{g}$.



We can extend $\tilde{g}$ further to a function $G : \mathbb{C} \setminus \{z = 0\} \rightarrow \mathbb{C}$ in the following way

$$G(z) = \begin{cases} \tilde{g}(z) & z \notin L \\ g(z) & z \in L \cap \mathbb{H} \\ \overline{g(\bar{z})} & z \in L \cap -\mathbb{H}. \end{cases}$$

The proof that Scharz reflection gives a holomorphic function also proves that this function is holomorphic as a function $\mathbb{C} \setminus \{z = 0\} \rightarrow \mathbb{C}$.

We can translate G into a holomorphic extension of f by conjugating $h^{-1} \circ G \circ h : \mathbb{C} \setminus \{p\} \rightarrow \mathbb{C}$ where p is the preimage of 1 under the continuous extension $\bar{f}$.

If f has more than one point on S^1 which is sent to 1 (i.e. more than one point on $\mathbb{R}$ is sent to ∞ under h) then we can repeat a similar procedure, except now we Schwarz reflect vertical strips bounded by the vertical lines through the points on the real line which are sent to ∞ by h .

Lastly, must there only be finitely many points?

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4. Find two bounded open subsets of $\mathbb{C}$ that are homeomorphic but not biholomorphic.

We know from the Riemann mapping theorem that any simply connected open set (which is not all of $\mathbb{C}$) is biholomorphic to the unit disc. And so, any two bounded simply connected open sets are biholomorphic to each other. Thus, we need at least one of our open sets to be non-simply connected. Moreover, since we want to find a homeomorphism between our open sets, it follows then that both are not simply connected (and must have the same homology).

Let us conjecture that two annuli of inner radius 1 and different outer radii are not biholomorphic. Let us denote these annuli by A_1, A_2 , the first having outer radius r_1 and the latter having outer radius r_2 . Suppose for contradiction that there exists a biholomorphism $f : A_1 \rightarrow A_2$.

Our initial biholomorphism induces a biholomorphism of vertical strips of differing width as follows. These have universal covers of the vertical strip in the complex plane, the universal cover S_1 for A_1 is the strip of width $\log r_1$ and the latter S_2 has width $\log r_2$, both have their left side aligned with the vertical line at $z = 0$ and the covering map for both of these is e^z on the appropriate domains. Notice that we have a map from the simply connected space $S_1 \rightarrow A_2$ via $f \circ (z \mapsto e^z)$, by the homotopy lifting property we have a map $S_1 \rightarrow S_2$ (with the appropriate commuting diagram) and likewise we have a map $S_2 \rightarrow S_1$. Now, given the commuting diagrams we get from the homotopy lifting property, we in fact have that the induced map $S_1 \rightarrow S_2$ is bijective. Moreover, since all the maps involved are holomorphic, the induced map $S_1 \rightarrow S_2$ is in fact a biholomorphism.

I think it might be easier to show that these two strips cannot be biholomorphic but I am not sure how to show this

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5. Interpret the function $\log z$ on $\mathbb{H}$ as a degenerate Schwarz-Christoffel map with image a degenerate polygon.

First recall that we have a well defined logarithm on the upper half plane (take the branch cut on any ray contained in the lower half plane) and so we can write $\log z$ suggestively as

$$\log z = \int_0^z \frac{1}{\zeta} d\zeta.$$

On the other hand, writing $\log z = \log(re^{i\theta}) = \ln r + i\theta$, and remembering that points on the upper half plane have $0 < \theta < \pi$, the image $\ln(\mathbb{H})$ is the infinite horizontal strip given by $\{z \in \mathbb{C} : 0 < \text{Im}(z) < \pi\}$. This “looks” like a polygon whose corners are occurring off at infinity.

More precisely, any radial ray from $z = 0$ at some angle θ is mapped to a horizontal line (“moving to the right”) at $\text{im}(z) = \theta$. Consider the image of the boundary rays under $\log z$. We have that the ray $(0, \infty)$ is mapped to the “lower” boundary component $\text{im}(z) = 0$. On the other hand, the ray $(0, -\infty)$ is mapped to the “upper” boundary component at $\text{im}(z) = \pi$.

If we want to think of this map as a degenerate Schwarz-Christoffel map, then we might want to think of what a “continuous extension” of $\log z$ to the boundary might mean (of course, this is a heuristic argument since $\log z$ is ill defined at $z = 0$). Approaching $z = 0$ from $+\infty$ gives a point “infinitely far to the left” on the line in the image $\text{im}(z) = 0$ whereas approaching $z = 0$ from $-\infty$ gives a point also “infinitely far to the left” but now on the line $\text{im}(z) = \pi$. If we want to think about a continuous extension, we can then think of these two boundary lines as meeting at infinity. Then we can think of this infinite strip as having a “corner” at $z = \infty$ in the image, or at $z = 0$ on the boundary of the upper half plane. And indeed, this is manifest in $\log z = \int_0^z 1/\zeta d\zeta$ since the corners in Schwarz-Christoffel maps are supposed to occur at the images of the denominator points. That is all to say, we can think of the image of $\log z$ applied to the upper half plane as a degenerate polygon, with one corner at $z = \infty$. If we want to extend our source domain

to include the point at infinity, then we can also think of the image of $\log(z)$ as having another corner “infinitely far to the right”, or at the image of ∞ .

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